

Sohan Lele

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EDUCATION

University of Pennsylvania (GRASP Lab)

MSE in Robotics | GPA: 3.77/4.00

Aug. 2025 – Dec. 2026

Philadelphia, PA

University of California, San Diego

BS in Mechanical Engineering, Specialization in Robotics and Controls | GPA: 3.69/4.00

Sep. 2021 – Jun. 2025

La Jolla, CA

EXPERIENCE

IRIX | tryirix.com

Feb. 2026 – Present

Co-Founder | Wearable Systems, Product Design, Sensor Fusion

San Francisco, CA

- Co-founded a wearable health startup, owning product and the technical roadmap for an AI coaching platform built on Apple Watch, WHOOP, Garmin, and Oura data
- Designed and built a smart-glasses coaching prototype end to end, pairing a vision-language model that reads plate loads with on-device IMU signal processing that counts reps in real time, validated on squats, deadlifts, and pull-ups
- Designed a multi-camera and wristband tracking system that fuses video and motion data with an Extended Kalman Filter, tracking lifts through occlusion and identifying users without facial recognition
- Leading pilot, partnership, and investment discussions with 24 Hour Fitness and 5+ boutique fitness operators; built the pricing model and unit economics behind the pitch

SoundImaging | MRI Headphones (UCSD Senior Design)

Jan. 2025 – Jun. 2025

Product Design Engineer | Medical Devices, Acoustics, Mechanical Design

La Jolla, CA

- Designed a pneumatic headphone system from MRI-safe, FDA-compliant materials, delivering clear audio at up to 130 dB
- Improved signal-to-noise ratio by ~45% by redesigning the transducer housing in SolidWorks with a tool-free push-and-twist lid and sealing every acoustic joint
- Selected 1 in. PVC tubing over a louder but rigid nylon option for kink resistance and easy routing around the scanner

Eversun Energy Inc.

Jul. 2024 – Sep. 2024

Mechanical Engineering Intern | Prototyping, Mechanical Design, Ergonomics, Fabrication

San Diego, CA

- Led design and fabrication of the Apollo X eTower alpha, a solar lighting tower, from concept to demo-ready hardware in 3 months
- Eliminated recurring cable fraying in the leg-deployment mechanism by tracing it to rotational wear and adapting an off-the-shelf lever to pull cables linearly; smooth deployment became a key selling point in investor demos
- Specified the gas struts and their lengths and selected leg materials based on the loads and outdoor conditions the tower had to withstand
- Applied design-for-assembly principles, authoring the step-by-step assembly sequence for the alpha unit before handoff

PROJECTS

Brain-Computer Interface | Finger Movement Decoding

Jan. 2026 – May 2026

Machine Learning Project | Signal Processing, Deep Learning, Neural Interfaces

Philadelphia, PA

- Built a deep learning pipeline decoding individual finger movements from ECoG brain signals, engineering the filtering and feature extraction for noisy neural data
- More than doubled decoding accuracy over a linear baseline ($r = 0.24$ to 0.54) by iterating from a regularized linear decoder to per-subject CNN and GRU ensembles trained across multiple seeds
- Diagnosed why ring-finger predictions lagged, tracing errors to shared tendons and overlapping brain activity between fingers

Steerable Needle Position Estimation | Multi-View Computer Vision

Aug. 2025 – Dec. 2025

Graduate Research Project, GRASP Lab | Medical Robotics, Computer Vision

Philadelphia, PA

- Designed a two-camera perception pipeline tracking the 3D tip of a steerable needle through reflective gel, where stereo depth failed
- Achieved ~7% relative error (2.5 mm) against a physics-based deformation model by combining a segmentation model (SAM2) with classical vision techniques
- Diagnosed and resolved failure modes from glare artifacts, skeleton branching, and camera frame synchronization

Autonomous Projectile-Launching Vehicle | Mobile Robotics

Mar. 2025 – Jun. 2025

Mechanical Lead, UCSD ECE/MAE 148 | Mechanical Design, ROS, Computer Vision

La Jolla, CA

- Led mechanical design of a ROS robot that navigates with stereo depth, tracks targets with vision, and aims a pan-tilt launcher
- Chose servo actuation over complex firing mechanisms for reliability; designed the turret for rigidity and fast disassembly
- Delivered a working prototype that performed reliably in live demos after pivoting from GPS to depth-based navigation

SKILLS

Mechanical Design & CAD: SolidWorks, Fusion 360, Rhino 7, ANSYS (FEA), Mechanism Design, Materials Selection, Ergonomics

Manufacturing & Prototyping: DFM/DFA, GD&T, Tolerance Analysis, Process Selection, 3D Printing (FDM, SLA), CNC Machining

Robotics & Perception: ROS, Computer Vision, Sensor Fusion, Signal Processing, Machine Learning

Programming: Python, MATLAB, Git